



Activity 1 - Hospital Patient Admission Analysis

PYTHON

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
import squarify

sns.set_style("whitegrid")
np.random.seed(1)

departments = ['Cardiology', 'Neurology', 'Pediatrics', 'Orthopedics', 'Emergency']
months = pd.date_range("2025-01-01", periods=12, freq='MS').strftime('%b')
base_load = {'Cardiology': 60, 'Neurology': 40, 'Pediatrics': 50, 'Orthopedics': 45, 'Emergency': 90}

rows = []
for dept in departments:
    for m in months:
        admissions = max(5, int(np.random.normal(base_load[dept], base_load[dept] * 0.2)))
        rows.append([dept, m, admissions])
df = pd.DataFrame(rows, columns=['Department', 'Month', 'Admissions'])
annual_totals = df.groupby('Department')['Admissions'].sum().sort_values(ascending=False)

plt.figure(figsize=(7, 5))
annual_totals.plot(kind='bar', color='steelblue')
plt.title("Total Annual Admissions by Department (Bar Chart)")
plt.xlabel("Department"); plt.ylabel("Total Admissions")
plt.tight_layout()
plt.show()

pivot = df.pivot(index='Month', columns='Department', values='Admissions').reindex(months)
pivot.plot(kind='bar', stacked=True, figsize=(9, 5), colormap='tab10')
plt.title("Monthly Admissions by Department (Stacked Bar Chart)")
plt.xlabel("Month"); plt.ylabel("Admissions")
plt.legend(title="Department", bbox_to_anchor=(1.02, 1), loc='upper left')
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
squarify.plot(sizes=annual_totals.values, label=[f'{d}\n{v}' for d, v in annual_totals.items()],
              color=sns.color_palette("Set2", len(annual_totals)))
plt.title("Share of Total Admissions by Department (Treemap)")
plt.axis('off')
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Department', y='Admissions', hue='Department', palette='Set3', legend=False)
plt.title("Monthly Admission Variability by Department (Box Plot)")
plt.xlabel("Department"); plt.ylabel("Monthly Admissions")
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
plt.hist(df['Admissions'], bins=15, color='mediumpurple', edgecolor='black')
plt.title("Distribution of Monthly Admission Counts (Histogram)")
plt.xlabel("Admissions per Month"); plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

print("Evaluation: see accompanying write-up for visualization justification.")
```

Output



R

Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

departments <- c('Cardiology','Neurology','Pediatrics','Orthopedics','Emergency')
months <- month.abb
base_load <- c(Cardiology=60, Neurology=40, Pediatrics=50, Orthopedics=45, Emergency=90)

df <- data.frame()
for (dept in departments) {
  admissions <- pmax(5, round(rnorm(12, base_load[[dept]], base_load[[dept]]*0.2)))
  df <- rbind(df, data.frame(Department=dept, Month=months, Admissions=admissions))
}
df$Month <- factor(df$Month, levels=months)

annual_totals <- df %>% group_by(Department) %>% summarise(Total=sum(Admissions)) %>%
  arrange(desc(Total))
annual_totals$Department <- factor(annual_totals$Department, levels=annual_totals$Department)

p1 <- ggplot(annual_totals, aes(x=Department, y=Total, fill=Department)) +
  geom_col(show.legend=FALSE) +
  labs(title="Total Annual Admissions by Department (Bar Chart)", y="Total Admissions") +
  theme_minimal()
print(p1)

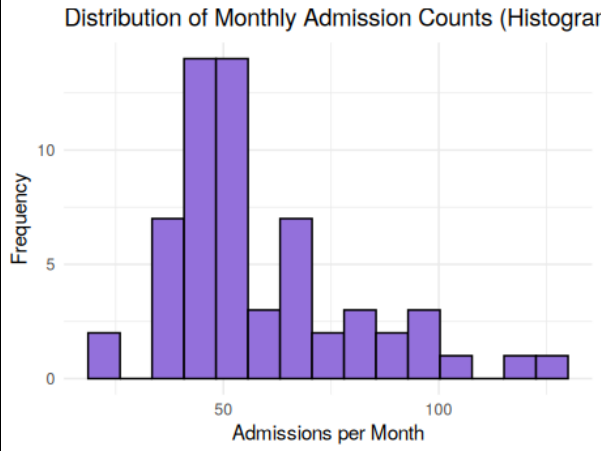
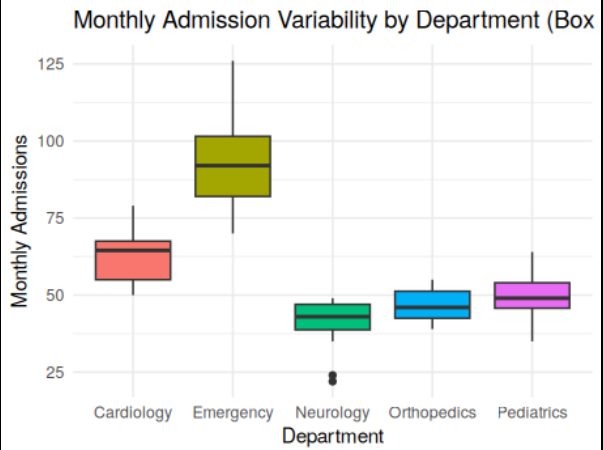
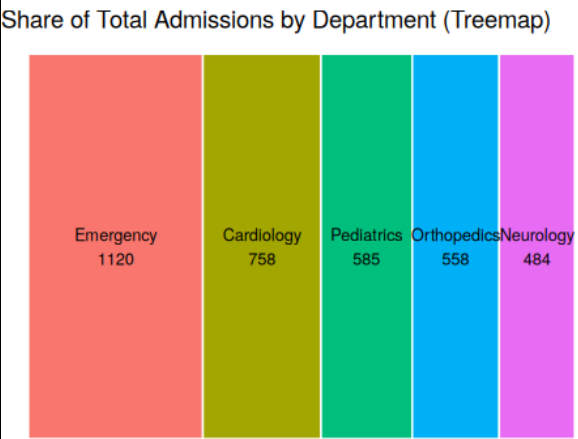
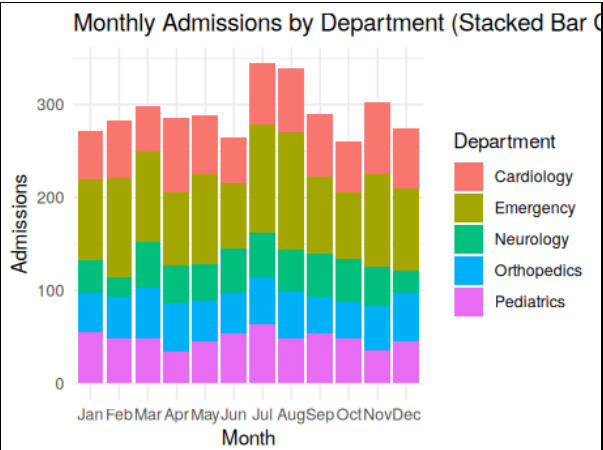
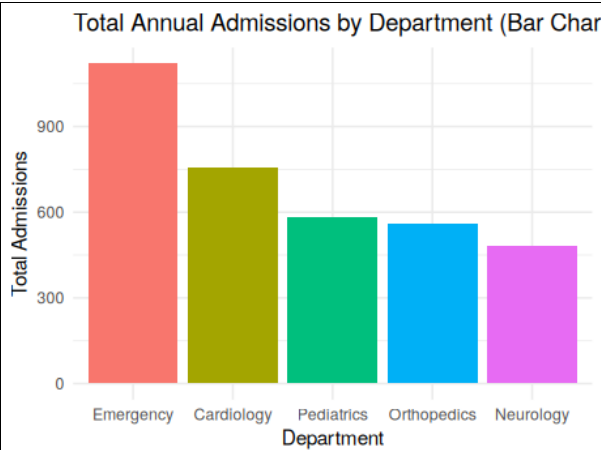
p2 <- ggplot(df, aes(x=Month, y=Admissions, fill=Department)) +
  geom_col() +
  labs(title="Monthly Admissions by Department (Stacked Bar Chart)", y="Admissions") +
  theme_minimal()
print(p2)

tm <- annual_totals %>% arrange(desc(Total)) %>%
  mutate(frac = Total / sum(Total), xmax = cumsum(frac), xmin = lag(xmax, default=0))
p3 <- ggplot(tm) +
  geom_rect(aes(xmin=xmin, xmax=xmax, ymin=0, ymax=1, fill=Department), color="white") +
  geom_text(aes(x=(xmin+xmax)/2, y=0.5, label=paste0(Department, "\n", Total)), size=3.2) +
  labs(title="Share of Total Admissions by Department (Treemap)") +
  theme_void() + theme(legend.position="none")
print(p3)

p4 <- ggplot(df, aes(x=Department, y=Admissions, fill=Department)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Monthly Admission Variability by Department (Box Plot)", y="Monthly Admissions") +
  theme_minimal()
print(p4)

p5 <- ggplot(df, aes(x=Admissions)) +
  geom_histogram(bins=15, fill="mediumpurple", color="black") +
  labs(title="Distribution of Monthly Admission Counts (Histogram)",
       x="Admissions per Month", y="Frequency") +
  theme_minimal()
print(p5)
```

Output



Activity 2 - E-Commerce Customer Spending Behaviour

PYTHON

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

n = 300
categories = ['Electronics', 'Fashion', 'Grocery', 'Home', 'Beauty']
df = pd.DataFrame({
    'CustomerID': range(1, n + 1),
    'SpendAmount': np.random.lognormal(mean=4.2, sigma=0.9, size=n).round(2),
    'OrderFrequency': np.random.poisson(6, n),
    'Category': np.random.choice(categories, n)
})

plt.figure(figsize=(7, 5))
plt.hist(df['SpendAmount'], bins=30, color='steelblue', edgecolor='black')
plt.title("Distribution of Customer Spend Amount (Histogram)")
plt.xlabel("Spend Amount ($)"); plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.violinplot(data=df, x='Category', y='SpendAmount', hue='Category', palette='Set2', legend=False)
plt.title("Spend Amount Distribution by Category (Violin Plot)")
plt.xlabel("Category"); plt.ylabel("Spend Amount ($)")
plt.tight_layout()
plt.show()

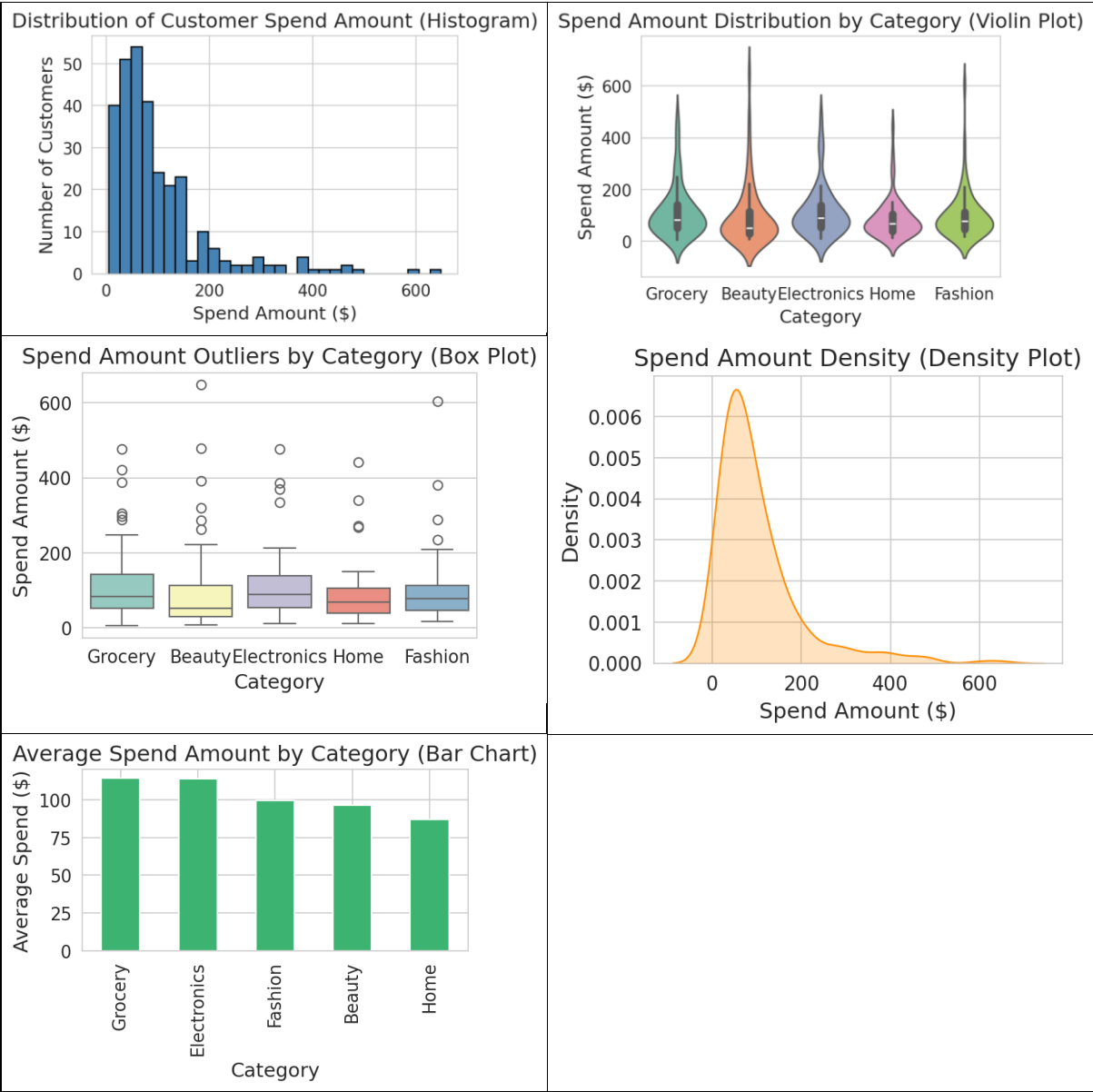
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Category', y='SpendAmount', hue='Category', palette='Set3', legend=False)
plt.title("Spend Amount Outliers by Category (Box Plot)")
plt.xlabel("Category"); plt.ylabel("Spend Amount ($)")
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
sns.kdeplot(df['SpendAmount'], fill=True, color='darkorange')
plt.title("Spend Amount Density (Density Plot)")
plt.xlabel("Spend Amount ($)"); plt.ylabel("Density")
plt.tight_layout()
plt.show()

avg_spend = df.groupby('Category')['SpendAmount'].mean().sort_values(ascending=False)
plt.figure(figsize=(7, 5))
avg_spend.plot(kind='bar', color='mediumseagreen')
plt.title("Average Spend Amount by Category (Bar Chart)")
plt.xlabel("Category"); plt.ylabel("Average Spend ($)")
plt.tight_layout()
plt.show()

print("Evaluation: see accompanying write-up for visualization justification.")
```

Output



R

Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

n <- 300
categories <- c('Electronics','Fashion','Grocery','Home','Beauty')
df <- data.frame(
  CustomerID = 1:n,
  SpendAmount = round(rlnorm(n, meanlog=4.2, sdlog=0.9), 2),
  OrderFrequency = rpois(n, 6),
  Category = sample(categories, n, replace=TRUE)
)

p1 <- ggplot(df, aes(x=SpendAmount)) +
  geom_histogram(bins=30, fill="steelblue", color="black") +
  labs(title="Distribution of Customer Spend Amount (Histogram)",
    x="Spend Amount ($)", y="Number of Customers") +
  theme_minimal()
print(p1)

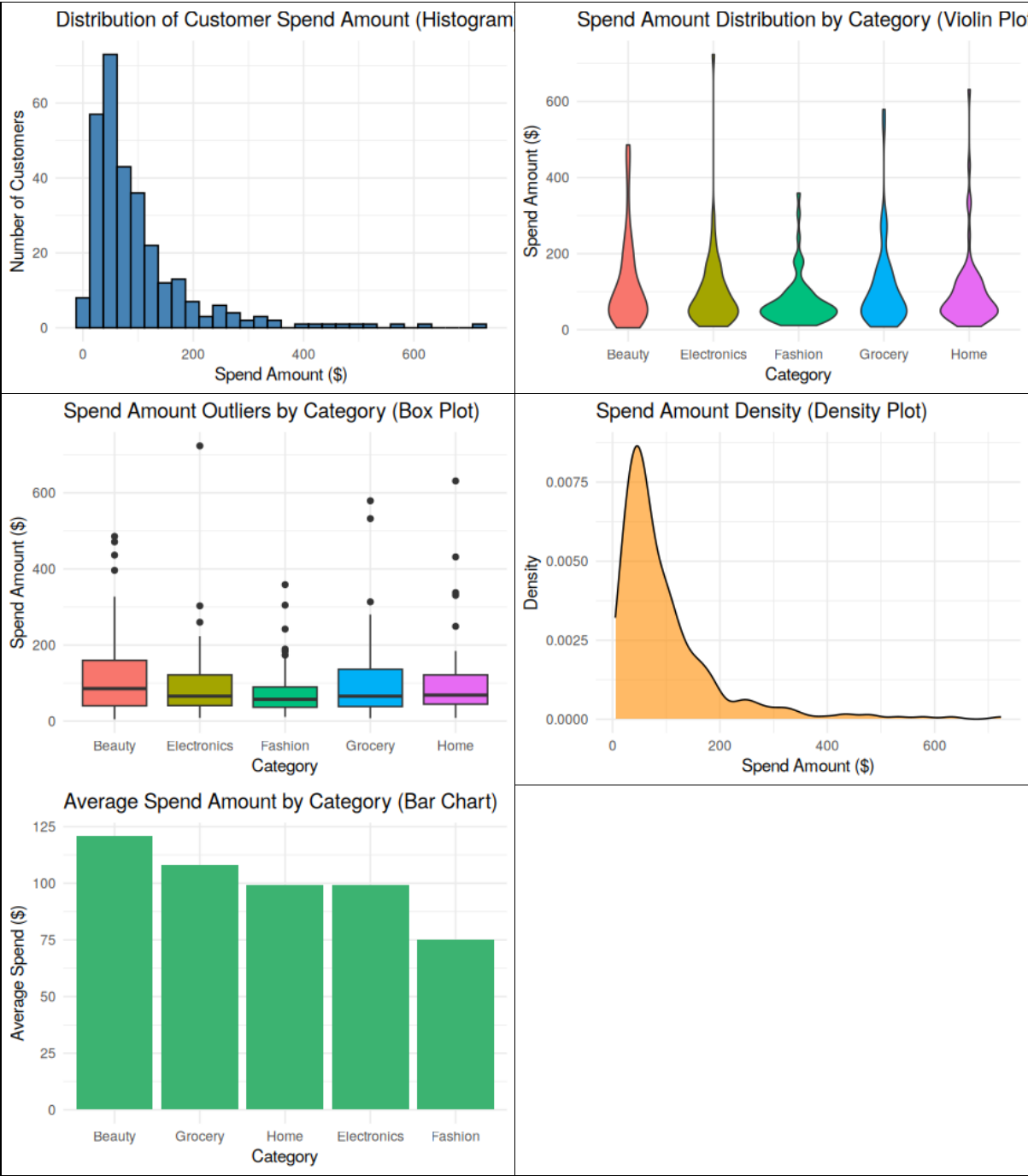
p2 <- ggplot(df, aes(x=Category, y=SpendAmount, fill=Category)) +
  geom_violin(show.legend=FALSE) +
  labs(title="Spend Amount Distribution by Category (Violin Plot)", y="Spend Amount ($)") +
  theme_minimal()
print(p2)

p3 <- ggplot(df, aes(x=Category, y=SpendAmount, fill=Category)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Spend Amount Outliers by Category (Box Plot)", y="Spend Amount ($)") +
  theme_minimal()
print(p3)

p4 <- ggplot(df, aes(x=SpendAmount)) +
  geom_density(fill="darkorange", alpha=0.6) +
  labs(title="Spend Amount Density (Density Plot)", x="Spend Amount ($)", y="Density") +
  theme_minimal()
print(p4)

avg_spend <- df %>% group_by(Category) %>% summarise(AvgSpend=mean(SpendAmount)) %>%
  arrange(desc(AvgSpend))
avg_spend$Category <- factor(avg_spend$Category, levels=avg_spend$Category)
p5 <- ggplot(avg_spend, aes(x=Category, y=AvgSpend)) +
  geom_col(fill="mediumseagreen") +
  labs(title="Average Spend Amount by Category (Bar Chart)", y="Average Spend ($)") +
  theme_minimal()
print(p5)
```


Output



Activity 3 - University Examination Performance Dashboard

PYTHON

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

departments = ['CSE', 'ECE', 'Mechanical', 'Civil', 'IT']
n_per_dept = 60
rows = []
dept_mean = {'CSE': 78, 'ECE': 72, 'Mechanical': 65, 'Civil': 68, 'IT': 80}
classes = ['Section A', 'Section B', 'Section C']

for dept in departments:
    for i in range(n_per_dept):
        marks = np.clip(np.random.normal(dept_mean[dept], 12), 0, 100)
        cls = np.random.choice(classes)
        rows.append([dept, cls, round(marks, 1)])
df = pd.DataFrame(rows, columns=['Department', 'Class', 'Marks'])

avg_marks = df.groupby('Department')['Marks'].mean().sort_values(ascending=False)
plt.figure(figsize=(7, 5))
avg_marks.plot(kind='bar', color='steelblue')
plt.title("Average Marks by Department (Bar Chart)")
plt.xlabel("Department"); plt.ylabel("Average Marks")
plt.tight_layout()
plt.show()

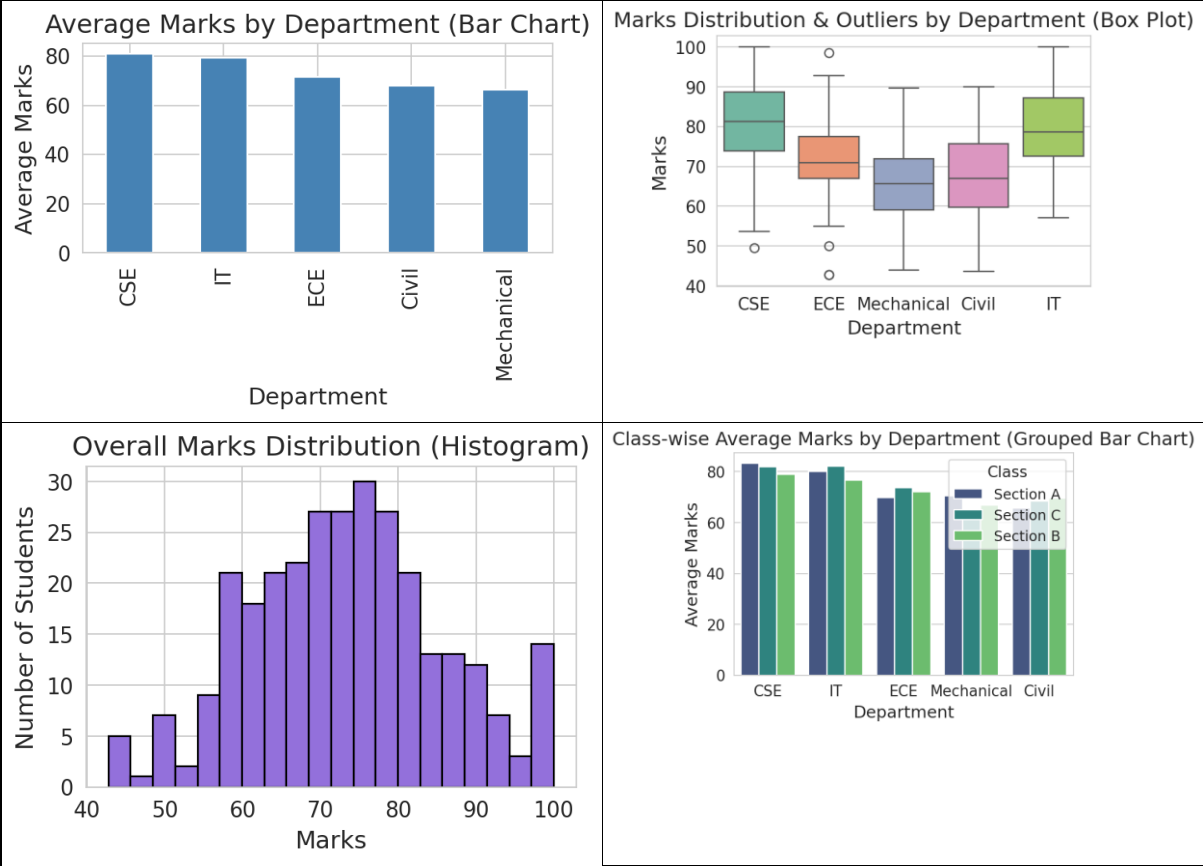
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Department', y='Marks', hue='Department', palette='Set2', legend=False)
plt.title("Marks Distribution & Outliers by Department (Box Plot)")
plt.xlabel("Department"); plt.ylabel("Marks")
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
plt.hist(df['Marks'], bins=20, color='mediumpurple', edgecolor='black')
plt.title("Overall Marks Distribution (Histogram)")
plt.xlabel("Marks"); plt.ylabel("Number of Students")
plt.tight_layout()
plt.show()

class_avg = df.groupby(['Department', 'Class', observed=True])['Marks'].mean().reset_index()
top_classes = class_avg.sort_values('Marks', ascending=False)
plt.figure(figsize=(9, 5))
sns.barplot(data=top_classes, x='Department', y='Marks', hue='Class', palette='viridis')
plt.title("Class-wise Average Marks by Department (Grouped Bar Chart)")
plt.xlabel("Department"); plt.ylabel("Average Marks")
plt.tight_layout()
plt.show()

print("Evaluation: see accompanying write-up for visualization justification.")
```

Output



R

Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

departments <- c('CSE','ECE','Mechanical','Civil','IT')
n_per_dept <- 60
dept_mean <- c(CSE=78, ECE=72, Mechanical=65, Civil=68, IT=80)
classes <- c('Section A','Section B','Section C')

df <- data.frame()
for (dept in departments) {
  marks <- pmin(pmax(rnorm(n_per_dept, dept_mean[[dept]], 12), 0), 100)
  cls <- sample(classes, n_per_dept, replace=TRUE)
  df <- rbind(df, data.frame(Department=dept, Class=cls, Marks=round(marks,1)))
}

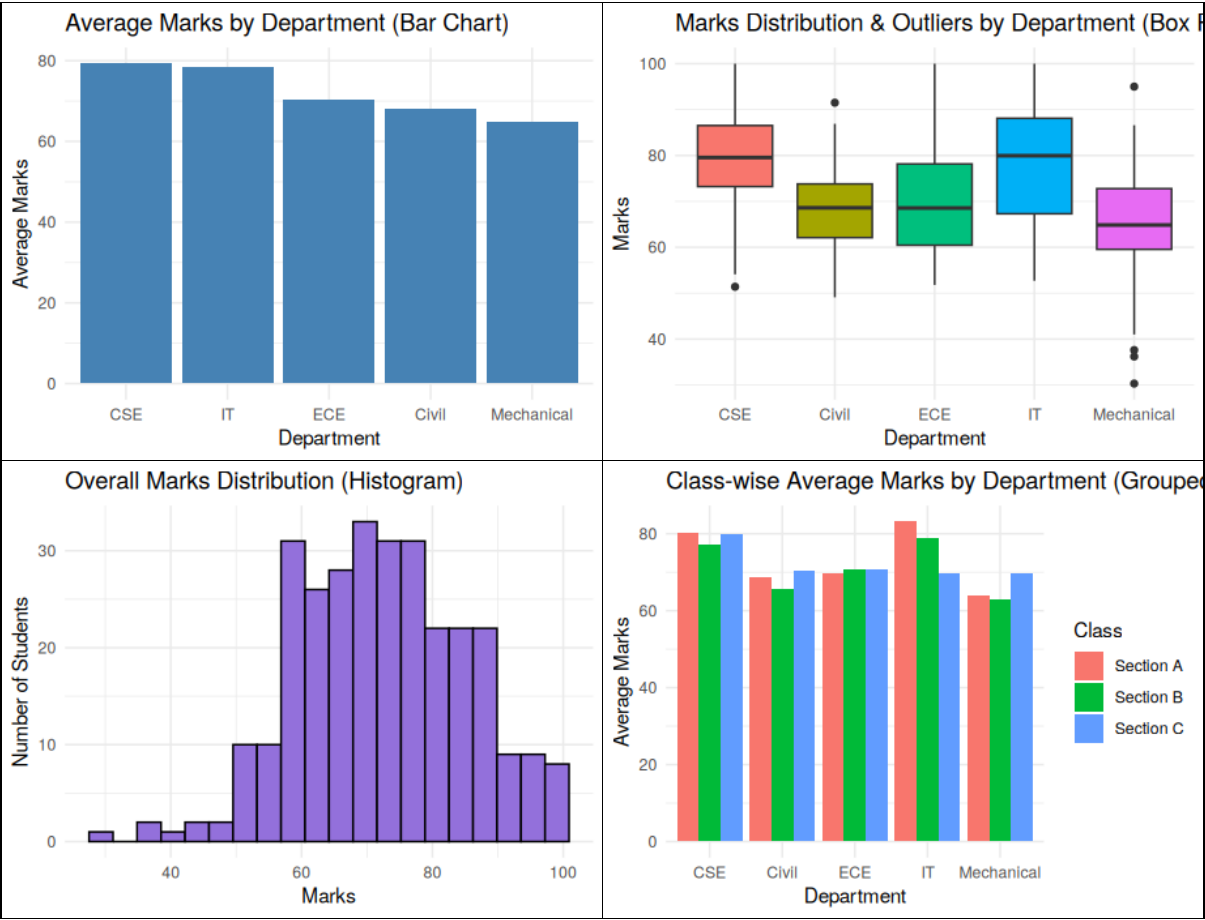
avg_marks <- df %>% group_by(Department) %>% summarise(AvgMarks=mean(Marks)) %>%
  arrange(desc(AvgMarks))
avg_marks$Department <- factor(avg_marks$Department, levels=avg_marks$Department)
p1 <- ggplot(avg_marks, aes(x=Department, y=AvgMarks)) +
  geom_col(fill="steelblue") +
  labs(title="Average Marks by Department (Bar Chart)", y="Average Marks") +
  theme_minimal()
print(p1)

p2 <- ggplot(df, aes(x=Department, y=Marks, fill=Department)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Marks Distribution & Outliers by Department (Box Plot)", y="Marks") +
  theme_minimal()
print(p2)

p3 <- ggplot(df, aes(x=Marks)) +
  geom_histogram(bins=20, fill="mediumpurple", color="black") +
  labs(title="Overall Marks Distribution (Histogram)", x="Marks", y="Number of Students") +
  theme_minimal()
print(p3)

class_avg <- df %>% group_by(Department, Class) %>% summarise(AvgMarks=mean(Marks), .groups='drop')
p4 <- ggplot(class_avg, aes(x=Department, y=AvgMarks, fill=Class)) +
  geom_col(position=position_dodge()) +
  labs(title="Class-wise Average Marks by Department (Grouped Bar Chart)", y="Average Marks") +
  theme_minimal()
print(p4)
```

Output



Activity 4 - Climate and Rainfall Distribution Study

PYTHON

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

cities = ['Chennai', 'Mumbai', 'Delhi', 'Bengaluru']
months = pd.date_range("2025-01-01", periods=12, freq='MS').strftime("%b")
monsoon_months = ['Jun', 'Jul', 'Aug', 'Sep']

rows = []
for city in cities:
    for m in months:
        base_rain = 180 if m in monsoon_months else 30
        rainfall = max(0, np.random.normal(base_rain, base_rain * 0.3))
        temp = np.random.normal(28, 4)
        humidity = np.random.normal(65, 10)
        rows.append([city, m, round(rainfall, 1), round(temp, 1), round(humidity, 1)])
df = pd.DataFrame(rows, columns=['City', 'Month', 'Rainfall', 'Temperature', 'Humidity'])

plt.figure(figsize=(7, 5))
plt.hist(df['Rainfall'], bins=20, color='steelblue', edgecolor='black')
plt.title("Distribution of Monthly Rainfall (Histogram)")
plt.xlabel("Rainfall (mm)"); plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

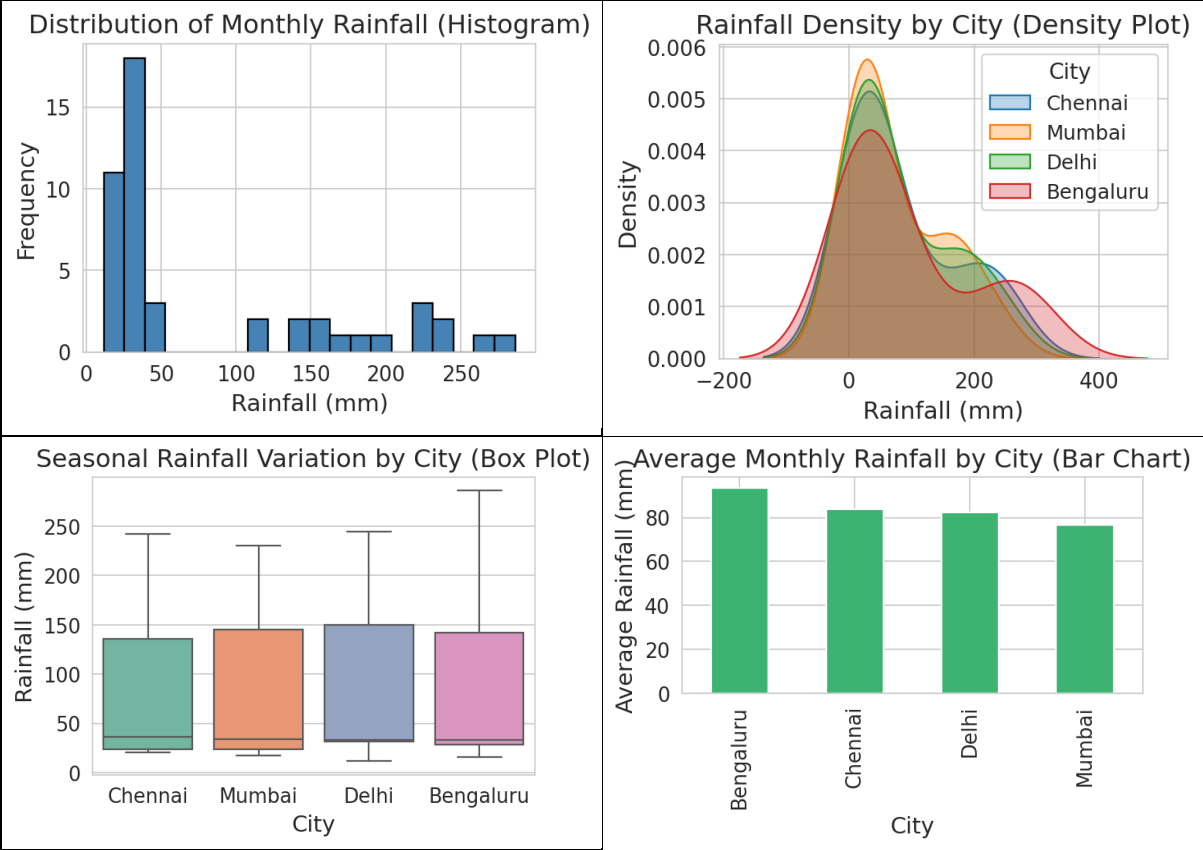
plt.figure(figsize=(8, 5))
for city in cities:
    sns.kdeplot(df[df.City == city]['Rainfall'], label=city, fill=True, alpha=0.3)
plt.title("Rainfall Density by City (Density Plot)")
plt.xlabel("Rainfall (mm)"); plt.ylabel("Density")
plt.legend(title="City")
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='City', y='Rainfall', hue='City', palette='Set2', legend=False)
plt.title("Seasonal Rainfall Variation by City (Box Plot)")
plt.xlabel("City"); plt.ylabel("Rainfall (mm)")
plt.tight_layout()
plt.show()

avg_rain = df.groupby('City')['Rainfall'].mean().sort_values(ascending=False)
plt.figure(figsize=(7, 5))
avg_rain.plot(kind='bar', color='mediumseagreen')
plt.title("Average Monthly Rainfall by City (Bar Chart)")
plt.xlabel("City"); plt.ylabel("Average Rainfall (mm)")
plt.tight_layout()
plt.show()

print("Evaluation: see accompanying write-up for visualization justification.")
```


Output



R

Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

cities <- c('Chennai','Mumbai','Delhi','Bengaluru')
months <- month.abb
monsoon_months <- c('Jun','Jul','Aug','Sep')

df <- data.frame()
for (city in cities) {
  for (m in months) {
    base_rain <- if (m %in% monsoon_months) 180 else 30
    rainfall <- max(0, rnorm(1, base_rain, base_rain*0.3))
    temp <- rnorm(1, 28, 4)
    humidity <- rnorm(1, 65, 10)
    df <- rbind(df, data.frame(City=city, Month=m, Rainfall=round(rainfall,1),
                             Temperature=round(temp,1), Humidity=round(humidity,1)))
  }
}

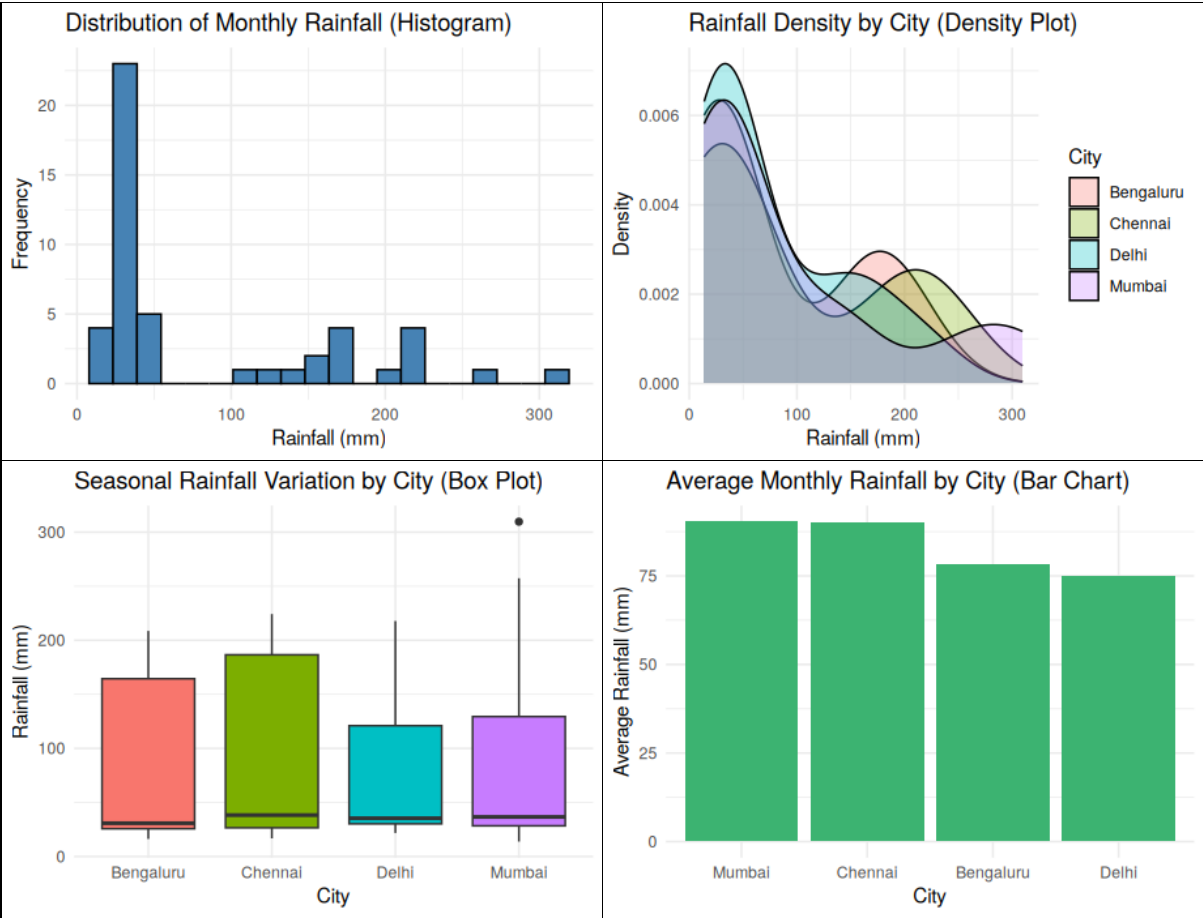
p1 <- ggplot(df, aes(x=Rainfall)) +
  geom_histogram(bins=20, fill="steelblue", color="black") +
  labs(title="Distribution of Monthly Rainfall (Histogram)", x="Rainfall (mm)", y="Frequency") +
  theme_minimal()
print(p1)

p2 <- ggplot(df, aes(x=Rainfall, fill=City)) +
  geom_density(alpha=0.3) +
  labs(title="Rainfall Density by City (Density Plot)", x="Rainfall (mm)", y="Density") +
  theme_minimal()
print(p2)

p3 <- ggplot(df, aes(x=City, y=Rainfall, fill=City)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Seasonal Rainfall Variation by City (Box Plot)", y="Rainfall (mm)") +
  theme_minimal()
print(p3)

avg_rain <- df %>% group_by(City) %>% summarise(AvgRainfall=mean(Rainfall)) %>%
  arrange(desc(AvgRainfall))
avg_rain$City <- factor(avg_rain$City, levels=avg_rain$City)
p4 <- ggplot(avg_rain, aes(x=City, y=AvgRainfall)) +
  geom_col(fill="mediumseagreen") +
  labs(title="Average Monthly Rainfall by City (Bar Chart)", y="Average Rainfall (mm)") +
  theme_minimal()
print(p4)
```

Output



Activity 5 - Bank Loan Risk Assessment

PYTHON

Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)
n = 300
df = pd.DataFrame({
    'CustomerID': range(1, n + 1),
    'LoanAmount': np.random.gamma(shape=3, scale=8000, size=n).round(0),
    'Income': np.random.normal(55000, 18000, n).clip(15000).round(0),
    'CreditScore': np.random.normal(680, 70, n).clip(300, 850).round(0),
})
df['RepaymentHistory'] = np.where(df['CreditScore'] > 650, 'Good',
                                  np.where(df['CreditScore'] > 550, 'Fair', 'Poor'))

plt.figure(figsize=(7, 5))
plt.hist(df['LoanAmount'], bins=25, color='steelblue', edgecolor='black')
plt.title("Distribution of Loan Amounts (Histogram)")
plt.xlabel("Loan Amount ($)"); plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
sns.boxplot(data=df, x='RepaymentHistory', y='LoanAmount',
            order=['Poor', 'Fair', 'Good'], hue='RepaymentHistory',
            palette='Set2', legend=False)
plt.title("Loan Amount by Repayment History (Box Plot)")
plt.xlabel("Repayment History"); plt.ylabel("Loan Amount ($)")
plt.tight_layout()
plt.show()

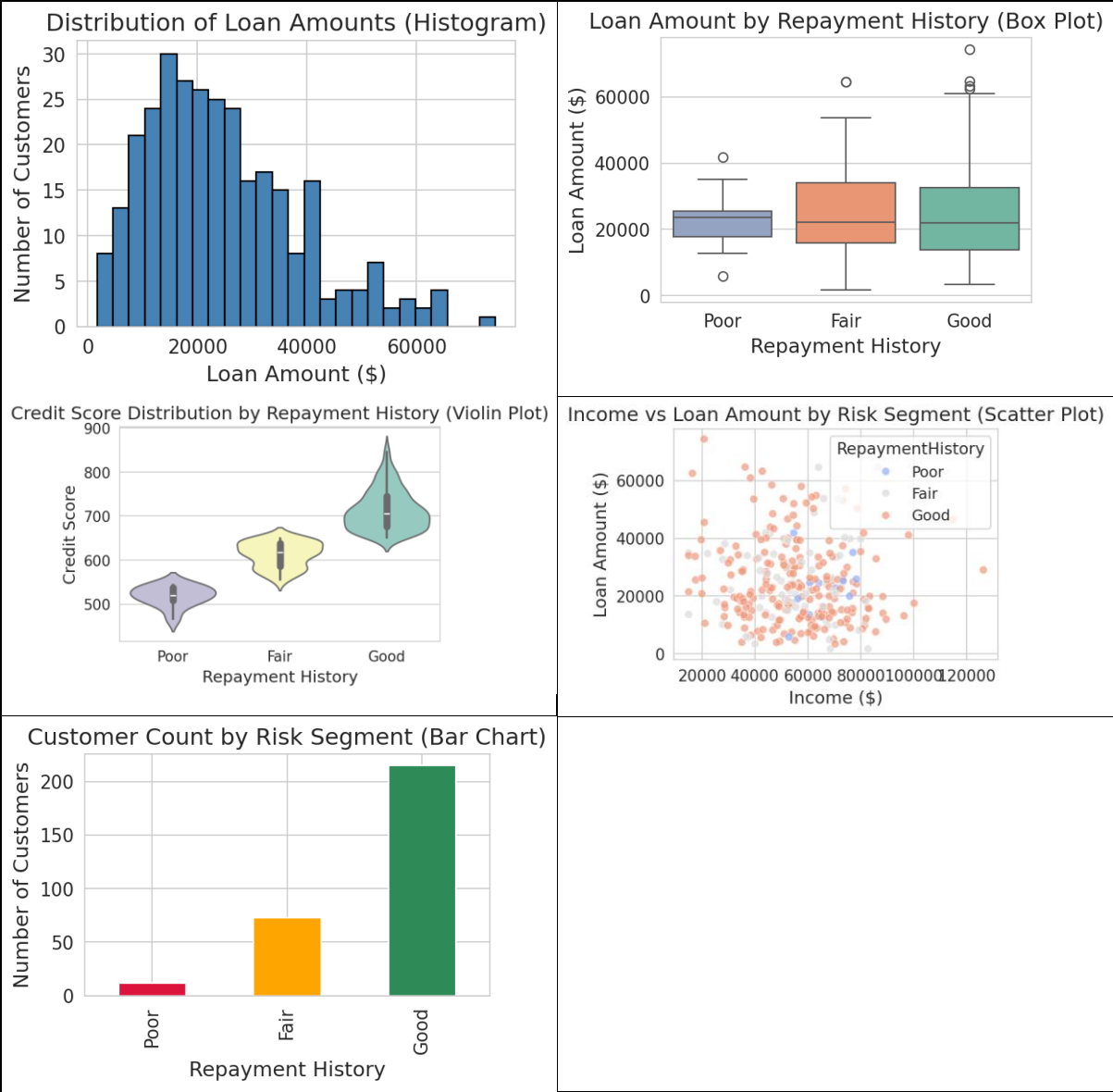
plt.figure(figsize=(7, 5))
sns.violinplot(data=df, x='RepaymentHistory', y='CreditScore',
               order=['Poor', 'Fair', 'Good'], hue='RepaymentHistory',
               palette='Set3', legend=False)
plt.title("Credit Score Distribution by Repayment History (Violin Plot)")
plt.xlabel("Repayment History"); plt.ylabel("Credit Score")
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
sns.scatterplot(data=df, x='Income', y='LoanAmount', hue='RepaymentHistory',
                hue_order=['Poor', 'Fair', 'Good'], palette='coolwarm', alpha=0.7)
plt.title("Income vs Loan Amount by Risk Segment (Scatter Plot)")
plt.xlabel("Income ($)"); plt.ylabel("Loan Amount ($)")
plt.tight_layout()
plt.show()

seg_counts = df['RepaymentHistory'].value_counts().reindex(['Poor', 'Fair', 'Good'])
plt.figure(figsize=(6, 5))
seg_counts.plot(kind='bar', color=['crimson', 'orange', 'seagreen'])
plt.title("Customer Count by Risk Segment (Bar Chart)")
plt.xlabel("Repayment History"); plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()

print("Evaluation: see accompanying write-up for visualization justification.")
```

Output



R

Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

n <- 300
df <- data.frame(
  CustomerID = 1:n,
  LoanAmount = round(rgamma(n, shape=3, scale=8000), 0),
  Income = round(pmax(rnorm(n, 55000, 18000), 15000), 0),
  CreditScore = round(pmin(pmax(rnorm(n, 680, 70), 300), 850), 0)
)
df$RepaymentHistory <- ifelse(df$CreditScore > 650, 'Good',
  ifelse(df$CreditScore > 550, 'Fair', 'Poor'))
df$RepaymentHistory <- factor(df$RepaymentHistory, levels=c('Poor','Fair','Good'))

p1 <- ggplot(df, aes(x=LoanAmount)) +
  geom_histogram(bins=25, fill="steelblue", color="black") +
  labs(title="Distribution of Loan Amounts (Histogram)", x="Loan Amount ($)", y="Number of Customers") +
  theme_minimal()
print(p1)

p2 <- ggplot(df, aes(x=RepaymentHistory, y=LoanAmount, fill=RepaymentHistory)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Loan Amount by Repayment History (Box Plot)", x="Repayment History", y="Loan Amount ($)")
+
  theme_minimal()
print(p2)

p3 <- ggplot(df, aes(x=RepaymentHistory, y=CreditScore, fill=RepaymentHistory)) +
  geom_violin(show.legend=FALSE) +
  labs(title="Credit Score Distribution by Repayment History (Violin Plot)",
    x="Repayment History", y="Credit Score") +
  theme_minimal()
print(p3)

p4 <- ggplot(df, aes(x=Income, y=LoanAmount, color=RepaymentHistory)) +
  geom_point(alpha=0.7) +
  scale_color_manual(values=c(Poor="firebrick", Fair="orange", Good="seagreen")) +
  labs(title="Income vs Loan Amount by Risk Segment (Scatter Plot)", x="Income ($)", y="Loan Amount ($)")
+
  theme_minimal()
print(p4)

seg_counts <- df %>% group_by(RepaymentHistory) %>% summarise(Count=n())
p5 <- ggplot(seg_counts, aes(x=RepaymentHistory, y=Count, fill=RepaymentHistory)) +
  geom_col(show.legend=FALSE) +
  scale_fill_manual(values=c(Poor="firebrick", Fair="orange", Good="seagreen")) +
  labs(title="Customer Count by Risk Segment (Bar Chart)", x="Repayment History", y="Number of
Customers") +
  theme_minimal()
print(p5)
```

Output

